

MODALEN III, Norway

WMO: 013260

Lat: 60.856N

Lon: 5.973E

Elev: 410

StdP: 14.48

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.3	13.0	4.1	6.9	15.5	6.8	7.9	16.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	76.9	61.6	72.5	59.8	68.7	57.7	63.4	72.1	61.6	68.8	59.8	66.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.6	80.3	64.9	58.7	74.8	63.0	56.9	70.2	61.5	29.0	71.5	27.7	69.1	26.4	66.3	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	DB	9.1	83.9	3.5	5.3	6.5	87.8	4.5	90.9	2.5	93.9	0.0	97.7	(4)
(5)			WB	7.2	65.2	4.3	2.2	4.1	66.8	1.6	68.1	-0.8	69.3	-3.9	70.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.9	30.0	29.6	33.9	40.7	47.8	53.7	59.1	57.3	51.5	43.4	36.0	30.8		
	DBStd	12.05	7.29	7.07	5.44	5.21	5.78	4.86	5.78	4.41	4.64	6.07	6.46	8.24		
	HDD50	3375	620	572	498	284	113	18	1	38	216	420	594			
	HDD65	8092	1085	992	963	730	534	341	200	243	406	670	869	1059		
	CDD50	783	0	0	0	4	45	128	285	226	82	12	1	0		
	CDD65	23	0	0	0	0	1	1	18	3	0	0	0	0		
	CDH74	262	0	0	0	0	9	32	199	22	0	0	0	0		
	CDH80	48	0	0	0	0	1	2	45	0	0	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	105.1	9.6	7.0	8.3	4.9	4.3	6.5	6.6	7.1	12.7	12.9	12.6	11.9		
(15) Precipitation	PrecMax	152.4	21.6	18.5	27.6	10.4	10.1	14.7	13.7	11.6	26.5	29.6	29.0	28.5		
	PrecMin	65.0	3.0	0.9	0.7	0.5	0.6	1.7	1.5	0.7	3.5	2.1	2.2	1.3		
	PrecStd	25.0	5.4	4.8	6.6	2.2	2.8	3.4	3.9	3.3	5.9	6.5	6.9	7.8		
	PrecStd	25.0	5.4	4.8	6.6	2.2	2.8	3.4	3.9	3.3	5.9	6.5	6.9	7.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.0	45.9	48.4	62.7	73.3	78.6	85.7	77.1	67.8	62.5	52.7	49.2		
		MCWB	41.6	40.5	41.0	50.2	56.7	60.0	65.0	62.7	57.3	54.4	47.9	45.6		
	2%	DB	42.8	42.6	44.8	58.7	67.9	72.3	79.3	72.7	64.3	56.6	49.8	45.5		
		MCWB	39.2	38.6	40.1	46.9	53.3	58.0	62.4	60.6	55.3	51.0	46.2	41.9		
	5%	DB	40.4	40.3	42.8	53.9	63.8	68.0	74.6	69.2	61.6	53.9	47.3	42.9		
		MCWB	37.3	37.3	39.0	44.5	51.3	54.6	61.3	59.3	55.2	50.5	44.1	40.1		
	10%	DB	38.3	38.1	40.9	49.6	59.2	64.2	70.3	65.6	58.9	51.6	44.8	40.7		
		MCWB	35.8	35.7	37.6	42.3	49.0	53.6	59.7	58.4	53.6	48.7	42.4	38.5		
	0.4%	WB	43.1	42.4	43.7	50.0	58.4	62.0	65.9	65.7	61.2	55.9	50.1	46.5		
		MCDB	44.7	44.6	46.1	58.3	72.1	73.1	79.7	71.1	63.2	57.5	51.5	47.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.1	39.7	41.4	46.9	54.7	59.3	63.9	63.2	58.3	53.1	47.3	42.9		
		MCDB	41.7	41.7	43.2	54.5	64.7	69.2	74.9	68.6	61.3	55.1	49.3	44.0		
	5%	WB	38.1	37.8	39.9	45.0	52.3	57.2	62.3	60.9	56.4	51.5	44.9	40.5		
		MCDB	39.8	39.6	41.7	50.6	60.9	64.6	71.1	65.8	59.5	53.4	46.4	42.1		
	10%	WB	36.4	36.1	38.3	43.5	50.3	55.2	60.6	59.2	54.8	49.4	42.5	38.4		
		MCDB	37.6	37.5	40.2	48.0	57.9	61.0	68.1	63.7	57.8	51.2	44.3	40.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	8.5	10.0	13.9	16.2	16.3	16.0	13.1	11.3	8.9	7.3	7.0		
		MCDBR	8.4	9.3	11.6	21.4	24.9	25.5	25.5	19.8	16.3	12.6	10.5	9.3		
		MCWBR	6.4	7.0	7.7	11.1	12.2	11.9	10.9	9.7	8.7	7.9	7.8	7.1		
	5% WB	MCDBR	8.1	8.4	10.2	17.2	22.0	21.1	22.5	15.8	12.8	10.1	10.1	8.9		
		MCWBR	6.7	7.0	7.9	9.9	11.3	10.7	10.2	8.9	8.5	7.6	8.4	7.5		
(40) Clear-Sky Solar Irradiance	taub	0.260	0.300	0.331	0.366	0.366	0.359	0.374	0.367	0.344	0.322	0.276	0.236			
	taud	2.082	2.183	2.183	2.178	2.348	2.420	2.407	2.421	2.476	2.414	2.184	1.999			
	Ebn at Noon	146	213	246	260	272	276	268	260	246	209	141	100			
	Edh at Noon	13	22	31	38	35	33	33	30	24	18	12	9			
(44) All-Sky Solar Radiation	RadAvg	69	230	590	1130	1532	1586	1465	1124	687	325	102	44			
	RadStd	14	54	105	156	187	186	153	112	90	49	16	6			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page